

Roderick Brown

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Education

University of California, Davis

M.S. Electrical Engineering

Sept 2026 - June 2027

- Research Topic: Highly Accurate mmWave Doppler Radar

B.S. Electrical Engineering - GPA: 3.8/4.0

Sept 2022 - June 2026

- Awards: ECE Citation Award, University Honors Program, Dean's Honors List
- Coursework: MOS Analog Circuit Design, RF Systems Design, Radiation & Antennas, IC Design & Tapeout

Related Experience

New Product Introduction Engineering Intern, Keysight Tech. – Santa Rosa, CA

June 2026 – Sept 2026

- Supported the development and validation of a 26-layer, 20 GHz signal source for Keysight's cutting-edge microwave signal generator.
- Isolated hardware failures to the component level in RF, analog, and digital domains by analyzing circuit schematics and assembly drawings and utilizing high-frequency test and measurement equipment.
- Collaborated with R&D engineers to develop solutions for permanent PCA modifications.

Systems Electrical Engineering Intern, Dept. of Viticulture & Enology – Davis, CA

June 2025 – Jan 2026

- Maintained and repaired Integrated Fermentation Control Systems (IFCS) units in winery lab to track important data such as temperature, brix, and pumpovers throughout fermentation process.
- Worked on IFCS RF communication, soldering 0402 resistors, and programming individual nodes.
- Developed a waste water monitor system with Click PLC and HMI. Complete with automated pump control, temperature, pH, and level sensors.

Projects

RF Power Amplifier Design & Tapeout

- Designed a 2.4 GHz class AB RF Power Amplifier with cascode driver stage in 180 nm CMOS
- Optimized transistor sizing to maximize output power using load-pull analysis and designed input, interstage, and output matching networks to optimize impedance matching.
- Designed and assembled an evaluation board to characterize and validate the performance of the taped-out IC.
- Tools Used: Cadence Virtuoso, KiCad, LT Spice, RF Signal Generator, Spectrum Analyzer, VNA

FMCW Radar

- Designed and characterized a 2.3 - 3.5 GHz FMCW radar as a part of 5 person team.
- Designed and assembled a baseband and VCO PCB to generate the transmit signal and amplify, filter, and process received and mixed signals
- Selected and integrated RF components to meet system-level input power requirements and ensure sufficient transmit power throughout the signal chain.
- Tools Used: Altium, LT Spice, VNA, Signal Generator, Spectrum Analyzer, Oscilloscope

Other Experience

Rec Sports Manager, Rec Sports at UC Davis – Davis, CA

Nov 2023 – Present

- Manage several rec sports leagues and club sport teams, overseeing and training a team of 25 staff members.
- Earned promotion through active program engagement, demonstrating leadership and strong communication skills.

Treasurer, Phi Delta Theta – Davis, CA

Dec. 2024 – Dec. 2025

- Analyzed financial data to make sustainable decisions and provided reports in weekly meetings.
- Created quarterly budgets and oversaw the cash flow of over \$150,000 annually, ensuring accurate accounting.

Skills

- Software: Altium, KiCad, HFSS, LT Spice, Cadence, Keysight ADS, Quartus, MATLAB, VS Code, Git, MS Office
- Electrical Debugging and Tools: Spectrum Analyzer, Network Analyzer, RF Signal Generator, DMM, Oscilloscope, Function Generator, Power Supply, Soldering (SMD & THT)
- Programming Languages: C/C++/C#, MATLAB, Java, Python, Perl, Verilog, Ladder logic